

Analysis and Design of Current Regulators Using Complex Vectors

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Abstract - The analysis and design of current regulators for multi-phase ac loads is presented using complex vector notation. The use of complex vector notation provides a way of comparing the performance of controller topologies through their complex vector root locus and complex vector frequency response functions. Limitations in the performance of the synchronous frame PI current regulator are outlined and several ways of improving its performance are suggested and investigated.

I. INTRODUCTION

The synchronous frame PI current regulator has become the standard for current regulation of multi-phase ac machines due to its capability of regulating ac signals over a wide frequency range [1, 2]. In a reference frame synchronous with the fundamental excitation, the fundamental excitation becomes a dc quantity which is easily regulated to the desired value using a PI controller. Even though the performance characteristics of the synchronous reference PI current regulator may seem intuitive, the multiple-input/multiple-output nature of the system makes performance evaluation difficult. In particular, the standard matrix or scalar notation does not easily lend itself to classical control design techniques such as root locus and frequency response function (FRF). The use of complex vector notation simplifies the model of an ac machine from a multiple-input/multiple-output system to an equivalent single-input/single-output complex vector system. The performance of this complex vector model can then be evaluated using generalized forms of the root locus and FRF, namely the complex vector root locus and complex vector FRF.

The synchronous frame PI current regulator can be modeled using both a nonlinear system representation [2] where the synchronous frequency, ω_e , is considered a system variable or as a linear system (Fig. 1) where ω_e is considered a time varying system parameter. Modeling ω_e as a parameter allows the use of linear systems tools with the realization that ω_e is assumed to be essentially constant.

This paper will systematically study the performance of synchronous reference frame current regulators. First the performance of the synchronous frame PI current regulator will be analyzed using scalar component notation. An RL load will be used initially due to its simple structure and reduced complexity. From this analysis it will become evident that the performance of this regulator is less than would be intuitively expected. Complex vector modeling will

then be introduced as a way of explaining the performance degradation and used to suggest several ways for improvement. Finally, the analysis for the RL load will be extended to the case of ac machine current control.

II. SCALAR ANALYSIS OF SYNCHRONOUS FRAME PI CURRENT REGULATORS

Using scalar notation the dq model for a symmetric, 3 phase RL load in the stationary reference frame is:

$$\begin{bmatrix} v_q^s \\ v_d^s \end{bmatrix} = \begin{bmatrix} R & 0 \\ 0 & R \end{bmatrix} \begin{bmatrix} i_q^s \\ i_d^s \end{bmatrix} + \begin{bmatrix} L & 0 \\ 0 & L \end{bmatrix} p \begin{bmatrix} i_q^s \\ i_d^s \end{bmatrix} \quad (1)$$

where p is the derivative operator.

Transforming this model to a reference frame synchronous with the fundamental excitation results in (2).

$$\begin{bmatrix} v_q^e \\ v_d^e \end{bmatrix} = \begin{bmatrix} R & \omega_e L \\ -\omega_e L & R \end{bmatrix} \begin{bmatrix} i_q^e \\ i_d^e \end{bmatrix} + \begin{bmatrix} L & 0 \\ 0 & L \end{bmatrix} p \begin{bmatrix} i_q^e \\ i_d^e \end{bmatrix} \quad (2)$$

If current control is desired a synchronous frame PI current regulator can be added to the RL load resulting in the block diagram shown in Fig. 1.

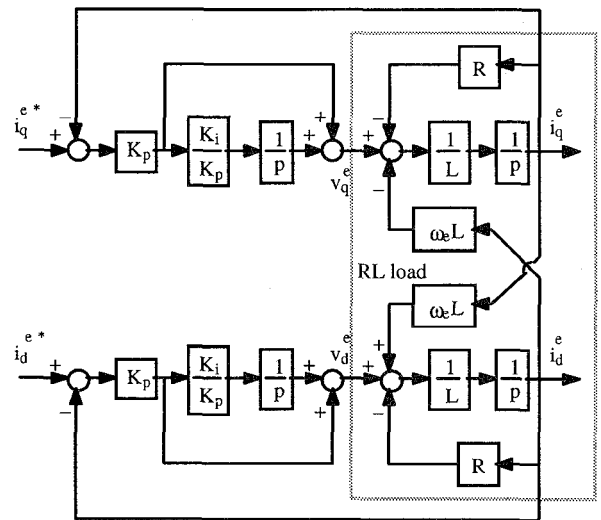


Figure 1: Scalar Block Diagram of an RL Load with a Synchronous Frame PI Current Regulator, Shown in the Synchronous Reference Frame

From (2) and Fig. 1 it can easily be seen that in the synchronous reference frame the RL load has two complex conjugate poles located at $-R/L \pm j\omega_e$. The classical

synchronous frame PI current regulator adds two open-loop poles to the system located at the origin (the integrators) and two real zeroes located at $-K_i/K_p$.

The performance of the classical synchronous frame PI current regulator was analyzed by applying it to a 3-phase RL load with the parameters shown in Table 1. The current regulator was tuned by selecting a controller zero approximately equal to the break frequency of the RL load, i.e., $K_i/K_p = R/L$. The controller gain was selected to achieve a relatively low bandwidth of 200 Hz which will exaggerate the control loop's inherent frequency dependency. It is readily acknowledged that a higher bandwidth would significantly improve the overall performance of the system.

An overlay of the commanded, theoretical, and experimental system response to step changes in the q and d axis current commands is shown in Fig. 2 for constant input synchronous frequencies of 50 and 200 Hz.

TABLE 1 – RL Load Parameters	
Parameter	Value
R	1.1 Ω
L	3.7 mH

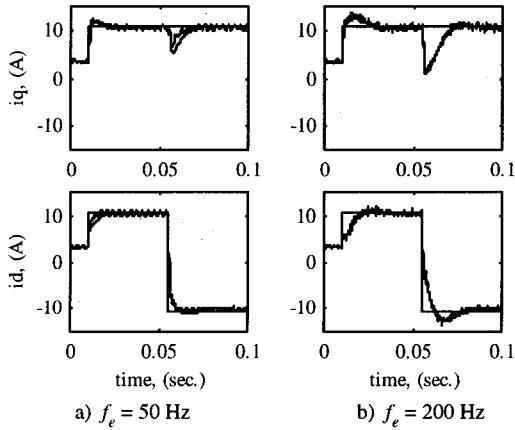


Figure 2: Overlay of the Theoretical, and Experimental q and d Axis Current Step Responses for an RL Load with a Synchronous Frame PI Current Regulator of 200 Hz bandwidth, Shown in the Synchronous Ref. Frame

From Fig. 2 it is readily apparent for a low synchronous frequency, 50 Hz, the system behaves nearly as expected for a 200 Hz current regulator bandwidth. For a high synchronous frequency, 200 Hz, serious degradation in the transient performance can be seen. The system responds with characteristics much closer to the time constant of the plant, L/R , than to the tuned bandwidth of 200 Hz. In addition, significant cross-coupling between the responses of the q and d axis currents can be seen.

The cause of this performance variation is the frequency dependent degradation of the pole/zero cancellation of the classical synchronous frame PI current regulator. As the synchronous frequency increases, the plant poles and controller zeroes migrate further and further apart reducing the overall system performance. Complex vector modeling

will now be used to gain substantial design insight into this frequency dependent performance degradation.

III. COMPLEX VECTOR MODELING

The dynamic models for polyphase ac systems can be represented using either complex vector or scalar notation [3 - 5]. The mapping of complex vector to scalar notation is shown in (3).

$$f_{qd} = f_q - j f_d \quad (3)$$

where f represents a generic complex vector quantity.

The complex vector itself can be represented in two forms: 1) cartesian, as shown in (3), and 2) polar as shown in (4).

$$f_{qd} = F e^{j\theta} \quad (4)$$

where $F = |f_{qd}|$ and $\theta = \text{atan}(-f_d/f_q)$.

Even though both scalar and complex vector representations give the same results, each has its advantages and disadvantages in analyzing a system. One of the advantages of using complex vectors is that they reduce the order of the system by one half. As a result, the complex vector model is often suitable for analysis using classical root locus, frequency response, transfer function, etc. methods.

IV. ANALYSIS OF SYNCHRONOUS FRAME PI CURRENT REGULATORS USING COMPLEX VECTORS

Using complex vector notation the model for an RL load in the synchronous reference frame is shown in (5).

$$v_{qd}^e = R i_{qd}^e + L (p + j \omega_e) i_{qd}^e \quad (5)$$

The order of the system has been reduced from a second-order to a first-order system. In addition, the system is now a single-input / single-output form, instead of the two input / two output form of the scalar model. The complex vector block diagram of the RL load with a synchronous frame PI current regulator is shown in Fig. 3.

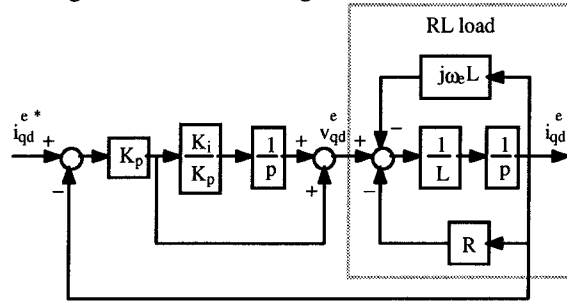


Figure 3: Complex Vector Block Diagram of an RL Load with a Synchronous Frame PI Current Regulator, Shown in the Synchronous Reference Frame

From (5) it is easily seen that in the synchronous reference frame the plant has a single, asymmetric, complex pole located at $-R/L - j\omega_e$. Tuning the current regulator identically to that done in the previous section ($K_i/K_p = R/L$, 200 Hz nominal bandwidth), the complex vector root locus can be plotted as shown in Fig. 4 for three different synchronous frequencies.

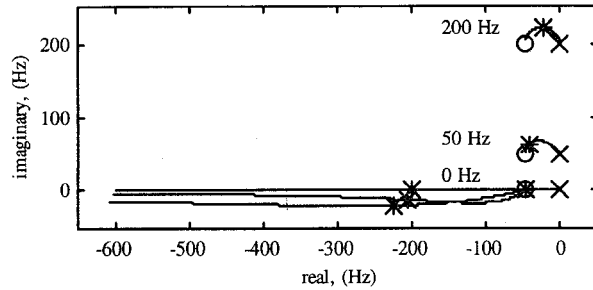


Figure 4: Complex Vector Root Locus of an RL Load with a Synchronous Frame PI Current Regulator (200 Hz bandwidth), Shown in the Stationary Reference Frame ($f_e = 0, 50$, and 200 Hz)

(O - zero, X - open loop pole, * - closed loop pole)

There are several key observations which can be made from the three root loci plots. First, the complex vector root locus is not symmetric about the real axis. This is due to the fact that the inputs and outputs are no longer real numbers but complex vectors and it is possible to get complex, asymmetric poles and zeroes. Second, the performance degradation of the synchronous frame PI current regulator as the synchronous frequency increases can be deduced as follows.

At low frequencies the controller zero (a real number zero) approximately cancels the plant pole (when the physical pole is mostly real). This allows the response of the system to be dominated by the faster closed loop pole, placed at the desired 200 Hz bandwidth.

For higher synchronous frequencies the controller zero interacts more with the integration pole added by the controller. The resulting slower root moves progressively closer to the imaginary axis away from the zero, so increasing overshoot is to be expected.

Because the system has a single-input / single-output form when modeled using complex vector notation a single transfer function completely describes its behavior, (6).

$$\frac{i_{qd}^s}{i_{qd}^{s*}} = \frac{K_p s + K_i - j K_p \omega_e}{L s^2 + (K_p + R - j \omega_e L) s + K_i - j \omega_e (K_p + R)} \quad (6)$$

The complex vector frequency response function (FRF), shown in Fig. 5, can be calculated from this transfer function. The complex vector FRF is plotted with both positive and negative frequencies since it is possible for complex vectors to rotate both forwards (positive frequencies) and backwards (negative frequencies). Note that this also implies that complex vector FRF's require a linear (rather than logarithmic) frequency scale. If the root locus were symmetric about the real axis the FRF would be identical for positive and negative frequencies, but because of the asymmetry shown in the root locus the FRF is also asymmetric for positive and negative frequencies.

From Fig. 5 it is apparent that the change in the complex vector FRF as the synchronous frequency increases is rather dramatic. The overshoot and undershoot as the synchronous frequency increases is as expected from the root locus above.

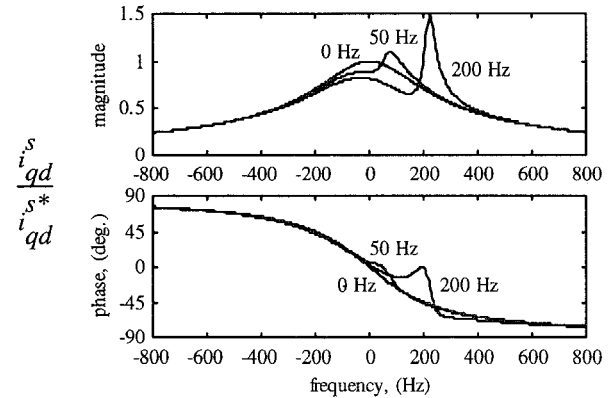


Figure 5: Complex Vector FRF of an RL Load with a Synchronous Frame PI Current Regulator (200 Hz bandwidth), Shown in the Stationary Reference Frame ($f_e = 0, 50$, and 200 Hz)

It is important to note that all of the FRF's shown in Fig. 5 have a unity gain and zero phase shift at the synchronous frequency. However, at frequencies away from the synchronous frequency there is significant distortion in the FRF.

It is important to understand the meaning of the FRF at frequencies other than the synchronous frequency. The synchronous frequency is the steady state fundamental component. Both disturbances and changes in the command trajectory will simultaneously excite the system with a wide range of frequency content. The FRF shows how the system will respond to the frequency content which is not at the synchronous frequency.

Complex vector notation can also be used to plot the time response of the system as the polar "magnitude" and polar vector angle ("phase") of the complex vector instead of as the magnitude of the q and d axis quantities individually. The time response shown in Fig. 2 is re-plotted in Fig. 6 using the complex vector magnitude and phase. The command trajectory for the time response can now be seen to consist of a step change in the magnitude command followed by a step change in the phase command.

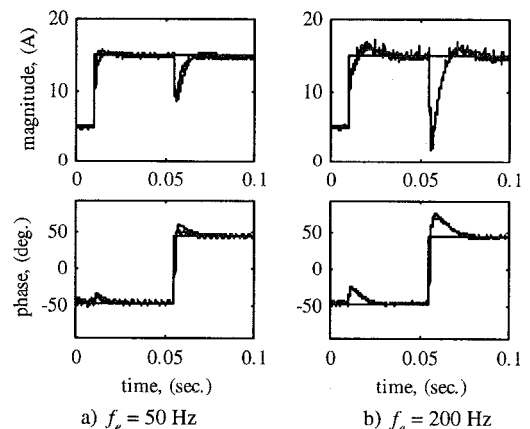


Figure 6: Overlay of Theoretical, and Experimental Current Magnitude and Phase Step Responses for an RL Load with a Synchronous Frame PI Current Regulator (200 Hz bandwidth), Shown in the Synchronous Reference Frame

Even though the commanded synchronous frequency remained constant for the time responses shown in Fig. 6, both the steps in the magnitude and phase of the commanded current introduce a large set of transient harmonics at frequencies centered around the synchronous frequency. Because of this, the transient response of the current regulator will depend on its capability to regulate frequencies other than the synchronous frequency. The FRF's shown in Fig. 5 are consistent with this behavior.

V. ALTERNATIVE DESIGN OF CURRENT REGULATORS FOR A 3 PHASE, SYMMETRIC RL LOAD

The deteriorated time response for the classical synchronous frame PI current regulator has been explained using complex vector FRF and root locus methods. An ideal synchronous reference frame current regulator would have a time response independent of the synchronous frequency when viewed in the synchronous reference frame. Such a regulator would have a complex vector FRF with a shape that does not vary with the synchronous frequency. Instead, the center of the FRF shape would just shift so that it is always symmetric about the synchronous frequency. With this in mind there are several ways of modifying the synchronous frame PI current regulator to achieve the desired results.

A. Synchronous Frame PI Current Regulator with Cross-Coupling Decoupling

One way of modifying the synchronous frame PI current regulator is to decouple the cross-coupling between the q and d axes caused by the $\omega_e L$ term in (2), which corresponds to decoupling $j\omega_e L$ in (5). Decoupling this term will make the performance of the current regulator independent of the synchronous frequency. The block diagram of the cross-coupling decoupling form of the synchronous frame PI current regulator is shown in Fig. 7.

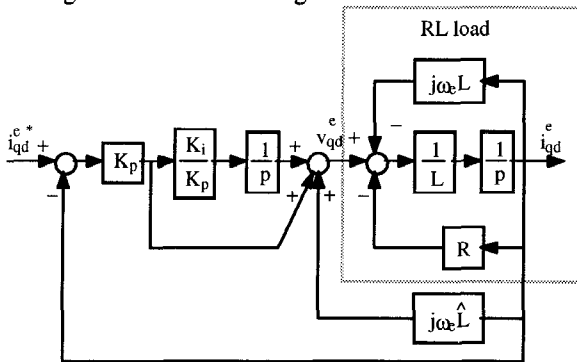


Figure 7: Complex Vector Block Diagram of an RL Load with a Cross-Coupling Decoupling Synchronous Frame PI Current Regulator, Shown in the Synchronous Reference Frame

The effect of the cross-coupling decoupling is to move the pole of the plant from $-R/L - j\omega_e$ to $-R/L$ in the synchronous reference frame, or from $-R/L$ to $-R/L + j\omega_e$ in the stationary reference frame. By moving the pole to this location it is now possible to directly cancel it using the real zero added by the

controller. In fact, the PI controller gains necessary to achieve this pole/zero cancellation and the desired bandwidth are identical to those used for the synchronous frame PI current regulator in the previous sections ($K_i/K_p = R/L$, with K_p tuned for the desired bandwidth). The resulting complex vector root locus for the cross-coupling decoupling synchronous frame PI current regulator is shown in Fig. 8 for three different synchronous frequencies.

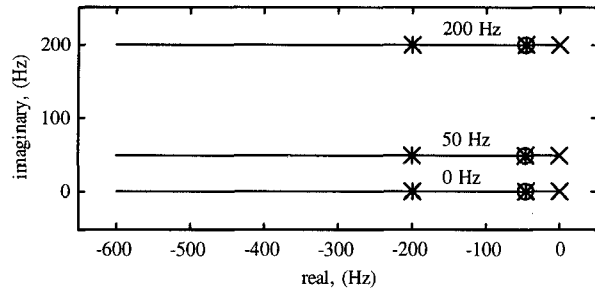


Figure 8: Complex Vector Root Locus of an RL Load with a Cross-Coupling Decoupling Synchronous Frame PI Current Regulator (200 Hz bandwidth), Shown in the Stationary Reference Frame ($f_e = 0, 50$, and 200 Hz) (O - zero, X - open loop pole, * - closed loop pole)

B. Complex Vector Synchronous Frame PI Current Regulator

The methodology used for the design of the cross-coupling decoupling synchronous frame PI current regulator suggests an alternative way of achieving similar results. The cross-coupling decoupling synchronous frame PI current regulator approximately moved the plant pole to the location of the controller zero through the use of decoupling. The dual to this methodology is to move the controller zero to the location of the plant pole by modifying the controller structure. The modified form of the synchronous frame PI current regulator which achieves this desired pole/zero cancellation is shown in Fig. 9. This form of the synchronous frame PI current regulator will be called the complex vector synchronous frame PI current regulator throughout the rest of the paper.

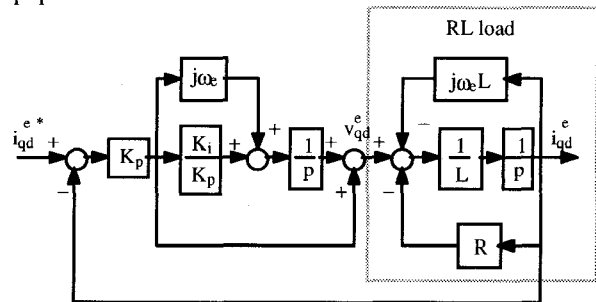


Figure 9: Complex Vector Block Diagram of an RL Load with a Complex Vector Synchronous Frame PI Current Regulator, Shown in the Synchronous Reference Frame

The basic design theory behind the complex vector synchronous frame PI current regulator is to place the zero added by the controller approximately on top of the plant

pole, even if the plant pole is complex. This is directly analogous to the classical control pole/zero cancellation methodology, with the only difference being the use of complex vectors which allows the placement of the controller zero off of the real axis. For the case of the RL load this results in the plant pole remaining at $-R/L - j\omega_e$ in the synchronous reference frame, or $-R/L$ in the stationary reference frame, with the controller zero placed on top of it. As was the case for the cross-coupling decoupling synchronous frame PI current regulator, the actual PI controller gains necessary to achieve this pole/zero cancellation and desired bandwidth are identical to the gains used for the synchronous frame PI current regulator in the previous sections ($K_i/K_p = R/L$, with K_p tuned for the desired bandwidth). The resulting complex vector root locus for the complex vector synchronous frame PI current regulator is shown in Fig. 10 for three different synchronous frequencies.

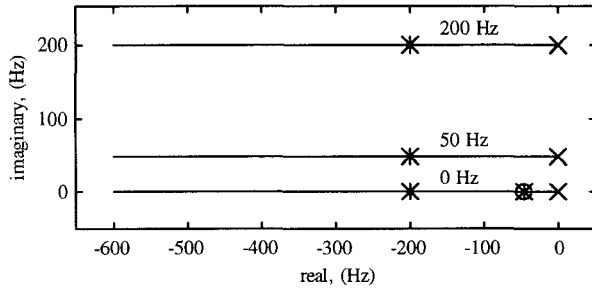


Figure 10: Complex Vector Root Locus for an RL Load with a Complex Vector Synchronous Frame PI Current Regulator (200 Hz bandwidth), Shown in the Stationary Reference Frame ($f_e = 0, 50$, and 200 Hz) (O - zero, X - open loop pole, * - closed loop pole)

The current regulators shown in Figs. 7 and 9 were implemented in the synchronous reference frame. Both have an equivalent stationary reference frame implementation which are not shown due to space constraints.

VI. ANALYSIS OF CURRENT REGULATORS FOR AN RL LOAD

As was done for the conventional synchronous frame PI current regulator it is possible to calculate the complex vector FRF's for the two modified forms of the synchronous frame PI current regulator. The transfer functions for the cross-coupling decoupling and complex vector forms are shown in (7) and (8) respectively.

$$\frac{i_{qd}^s}{i_{qd}^{s*}} = \frac{K_p s + K_i - j K_p \omega_e}{\left(L s^2 + (K_p + R - j \omega_e (L + \hat{L})) s + K_i - \omega_e^2 \hat{L} - j \omega_e (K_p + R) \right)} \quad (7)$$

$$\frac{i_{qd}^s}{i_{qd}^{s*}} = \frac{K_p s + K_i}{L s^2 + (K_p + R - j \omega_e L) s + K_i - j \omega_e R} \quad (8)$$

Although it isn't obvious from their transfer functions, if the parameter estimates are correct, and both current regulators

have the same PI gains, the complex vector FRF's for the two current regulators are identical and shown in Fig. 11.

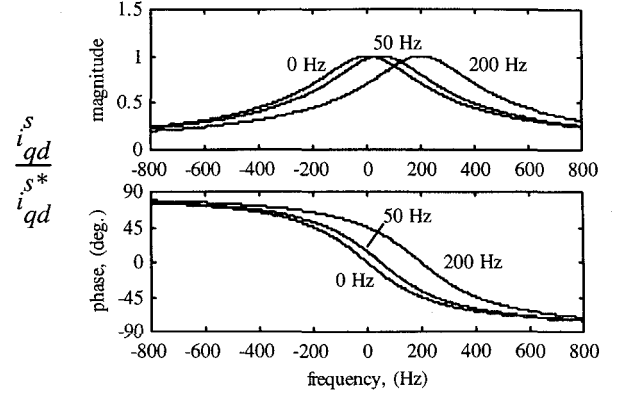


Figure 11: Complex Vector FRF of an RL Load for either a Cross-Coupling Decoupling or Complex Vector Synchronous Frame PI Current Regulator (200 Hz bandwidth), Shown in the Stationary Reference Frame ($f_e = 0, 50$, and 200 Hz)

The shape of the complex vector FRF shown in Fig. 11 is independent of the synchronous frequency and symmetric with respect to it. This is identical to the desired FRF.

Figure 12 shows an overlay of the theoretical and experimental step responses for the complex vector synchronous frame PI current regulator. (The cross-coupling decoupling controller has nearly identical characteristics and thus is not shown).

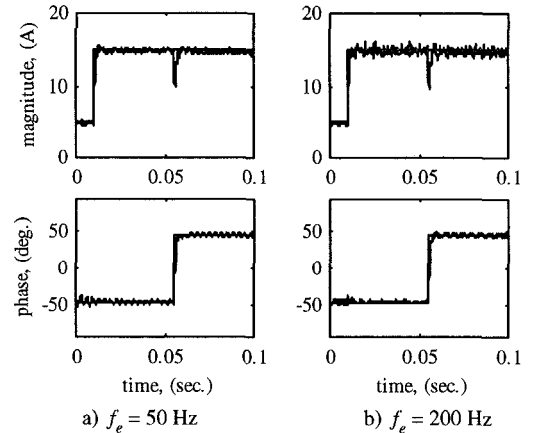


Figure 12: Overlay of the Theoretical, and Experimental Current Magnitude and Phase Step Response for an RL Load with a Cross-Coupling Decoupling & Complex Vector Synchronous Frame PI Current Regulators (200 Hz bandwidth), Shown in the Synchronous Reference Frame

The command trajectory goes through the same magnitude and phase variations used to test the conventional synchronous frame PI current regulator in Figs. 2 and 6. The improved performance exhibited by both of the modified synchronous frame PI current regulators is obvious. Their time responses correspond to the tuned bandwidth independent of the synchronous frequency.

The analysis of the synchronous frame PI current regulators heretofore has assumed that all of the parameters estimates used for tuning are identical to the actual system parameters.

A suitable parameter sensitivity analysis will now be presented. It will be limited to the two modified forms of the synchronous frame PI current regulator with the classical synchronous frame PI as benchmark.

Fig. 13 shows the time response for the cross-coupling decoupling and complex vector forms of the synchronous frame PI current regulator, respectively, for 20% underestimation of L at a synchronous frequency of 200 Hz. Although both current regulators still achieve performance significantly better than the conventional synchronous frame PI current regulator with correct parameter estimates, Fig. 13 shows that the complex vector form of the synchronous frame PI current regulator is less sensitive to errors in L .

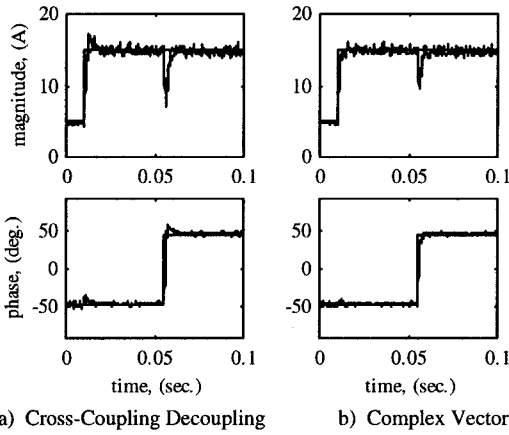


Figure 13: Overlay of the Theoretical, and Experimental Current Magnitude and Phase Step Responses for 20% Underestimation of L for an RL Load with Two Forms of Synchronous Frame PI Current Regulators (200 Hz bandwidth), Shown in the Synchronous Reference Frame ($f_e = 200$ Hz)

The reasons for the decreased parameter sensitivity of the complex vector form of the synchronous frame PI current regulator can be explained using the complex vector root locus. The complex vector synchronous frame PI current regulator achieves the desired pole/zero cancellation by moving the controller zero to the location of the plant pole. Because of this, the approximate pole/zero cancellation always occurs along the real axis of the root locus in the stationary reference frame. When the parameter estimates are incorrect the dynamics of any inexact pole/zero cancellation will be shifted away from the synchronous frequency for any synchronous frequency except dc. The cross-coupling decoupling synchronous frame PI current regulator, on the other hand, achieves the desired pole/zero cancellation by moving the plant pole to the controller zero. This forces the dynamics of any inexact pole/zero cancellation to move with the synchronous frequency and causes the resulting closed loop pole to move away from the zero with frequency.

The complex vector FRF can also be used to analyze the parameter sensitivity of the current regulators. Figs. 14 and 15 show the complex vector FRF's at a synchronous frequency of 200 Hz for errors in the inductance estimate of the cross-coupling decoupling and complex vector synchronous frame PI current regulators, respectively.

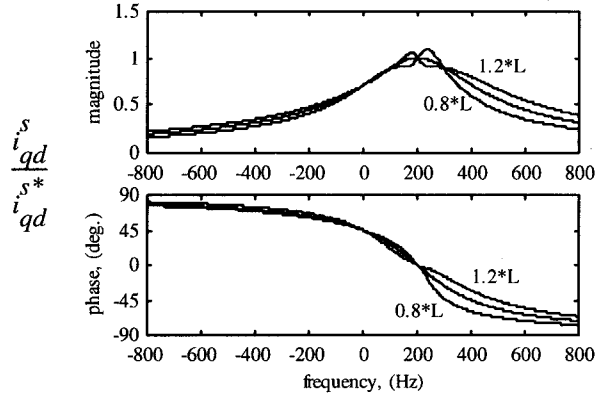


Figure 14: Complex Vector FRF for Variations in L of an RL Load with a Cross-Coupling Decoupling Synchronous Frame PI Current Regulator (200 Hz bandwidth), Shown in the Stationary Reference Frame ($f_e = 200$ Hz)

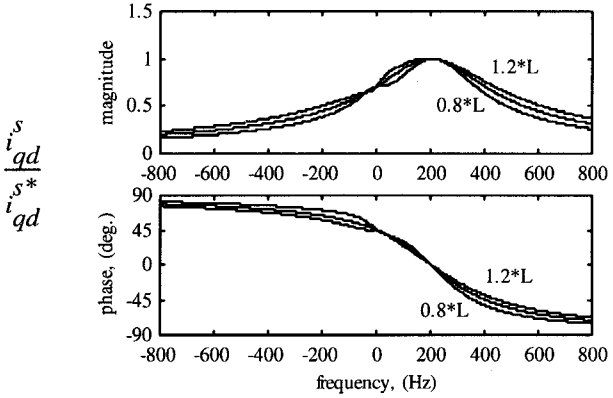


Figure 15: Complex Vector FRF for Variations in L of an RL Load with a Complex Vector Synchronous Frame PI Current Regulator (200 Hz bandwidth), Shown in the Stationary Reference Frame ($f_e = 200$ Hz)

The cross-coupling decoupling synchronous frame PI current regulator has the most significant deviations due to parameter errors centered right at the synchronous frequency, while the complex vector synchronous frame PI current regulator shifts these deviations away from the synchronous frequency.

VII. DESIGN OF CURRENT REGULATORS FOR AC MACHINES

An RL load was used for the initial discussion of complex vector design methods for synchronous frame current regulators. While this model and method may be appropriate for active filters and UPS systems, it will now be extended to the control of current for AC machine drives.

The equations governing the dynamic behavior of an induction motor with the stator current and rotor flux as the state variables are given in (9) and (10).

$$p i_{qds}^s = \frac{1}{L_{\sigma s}} (v_{qds}^s - r_s' i_{qds}^s + \frac{L_m}{L_r} \omega_{br} \lambda_{qdr}^s) \quad (9)$$

$$p \lambda_{qdr}^s = \frac{L_m}{L_r} r_r i_{qds}^s - \omega_{br} \lambda_{qdr}^s \quad (10)$$

$$\text{where } L_{\sigma s} = L_s - \frac{L_m^2}{L_r} \quad r_s' = r_s + \left(\frac{L_m}{L_r}\right)^2 r_r \quad \omega_{br} = \frac{r_r}{L_r} - j \omega_r$$

From these equations it is evident that the induction motor can be transformed into an equivalent RL load by approximately decoupling the back-emf caused by the rotor flux from the stator equation, (9), as shown in Fig. 16.

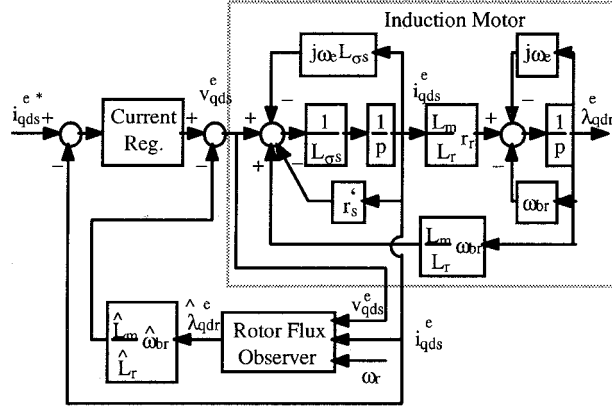


Figure 16: Complex Vector Block Diagram Showing the Implementation of a Current Regulator for a 3 Phase Induction Motor using Back-emf Decoupling, Shown in the Synchronous Reference Frame

Such approximate decoupling of the back-emf requires an estimate of the rotor flux such as in [4]. The accuracy of the rotor flux estimate used in the back-emf decoupling will affect the exact results, however, reasonable engineering precision ($\pm 10\%$) will achieve substantially the same results as was found in controlling the three phase, symmetric RL load. Thus, only the classical synchronous frame PI and the complex vector synchronous frame PI will be shown here.

In addition to the two cases which are nearly equivalent to the RL load, a more common case, where back-emf decoupling is not included in the controller, will be presented. The block diagram for this topology is shown in Fig. 17.

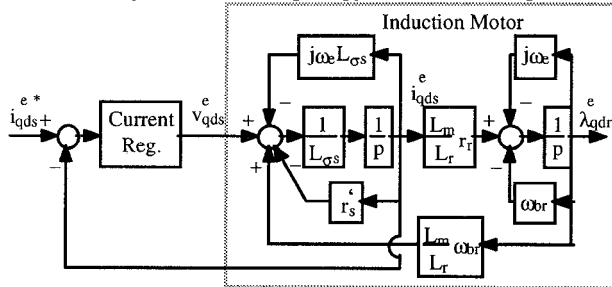


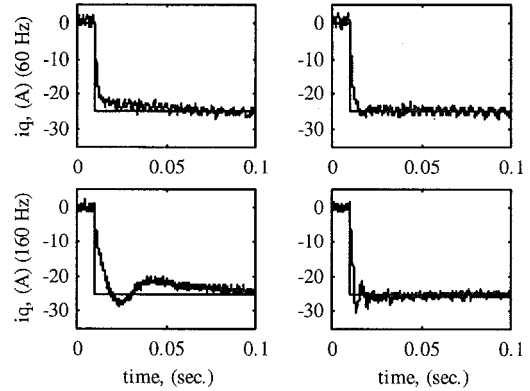
Figure 17: Complex Vector Block Diagram Showing the Implementation of a Current Regulator for a 3 Phase Induction Motor without Explicit Back-emf Decoupling, Shown in the Synchronous Reference Frame

All of the AC machine current regulator topologies in this paper were tuned for a 200 Hz bandwidth using an induction motor whose parameters are shown in Table 2.

A. Results: Current Regulators with Back-Emf Decoupling

Fig. 18 shows the experimental step response of the q-axis current for two different initial synchronous frequencies, 60 and 160 Hz. The improved performance of the complex vector current regulator at both low and high synchronous frequencies is quite apparent. The results are very similar to those for the RL load with the corresponding controllers.

TABLE 2 – Induction Motor Parameters	
Parameter	Value
r'_s	0.82 Ω
$L_{\sigma s}$	5.5 mH
L_m	148 mH
L_r	150 mH
r_r	0.36 Ω
Power Rating	6 KW
Poles	4



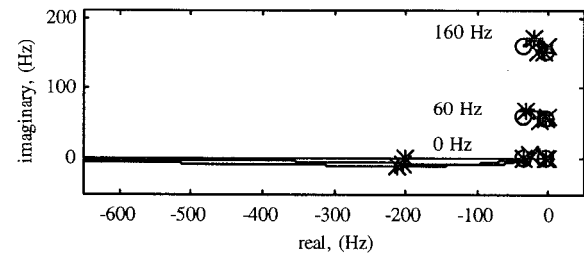
a) Classical, Synch. Ref. Frame PI with Decoupling b) Complex Vector Synch. Ref. Frame PI with Decoupling

Figure 18: Experimental q Axis Current Step Response for an Induction Motor with Classical and Complex Vector Synchronous Frame PI Current Regulator (200 Hz bandwidth) for $f_s = 60$ and 160 Hz, Shown in the Synchronous Reference Frame

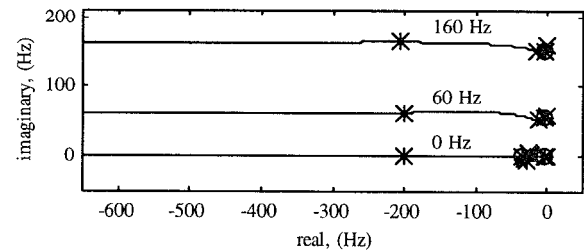
The results are very similar to those for the RL load with the corresponding controllers.

B. Results: Current Regulators without Back-Emf Decoupling

The complex vector root locus and FRF for both classical and complex vector synchronous frame current regulators without back-emf decoupling are shown in Figs. 19 and 20.

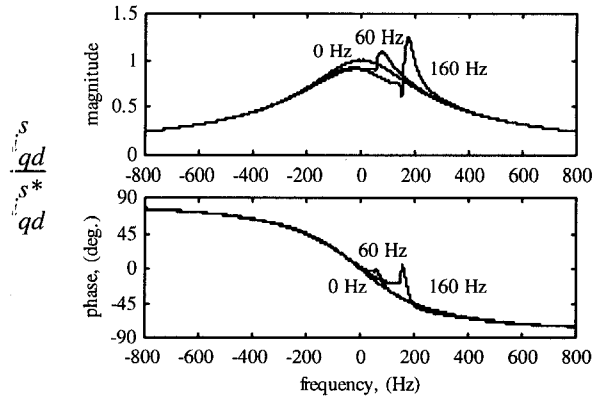


a) Classical Synchronous Frame PI Current Regulator without Decoupling

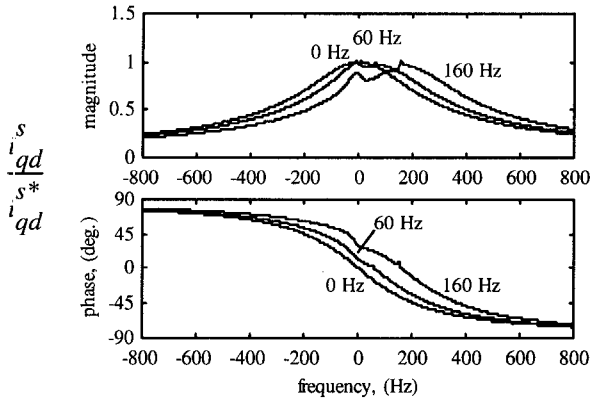


b) Complex Vector Synchronous Frame PI Current Regulator w/o Decoupling

Figure 19: Current Regulated Induction Motor Root Locus at Rated Slip (200 Hz bandwidth), Shown in the Stationary Reference Frame ($f'_c = 0, 60$ and 160 Hz) (O - zero, X - open loop pole, * - closed loop pole)



a) Classical Synchronous Frame PI FRF without Back-emf Decoupling



b) Complex Vector Synchronous Frame PI without Back-emf Decoupling
Figure 20: Complex Vector FRF at Rated Slip for an Induction Motor with Classical and Complex Vector Synchronous Frame PI Current Regulator (without Decoupling), 200 Hz bandwidth, for $f_e = 0, 60$, and 160 Hz, Shown in the Stationary Reference Frame

From the FRF it is apparent that the classical synchronous frame PI controller will have inferior dynamics at high synchronous frequencies as compared to the complex vector PI. The root loci provides insight into this degradation by showing how the closed loop root migrates from the zero at high frequencies causing overshooting properties in the FRF.

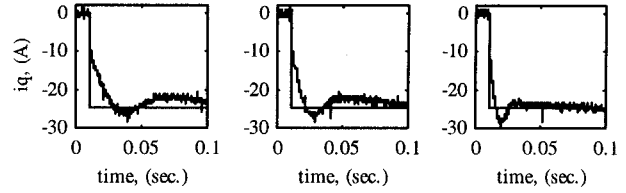
The comparative experimental step responses for the classical synchronous frame PI and the complex vector PI without back-emf decoupling are shown in Fig. 21 for a 160 Hz operating frequency. The results are shown for three different control zeros. The experimental step responses for all of the tuning variants show the dramatic potential improvement which can be achieved by use of the complex vector design as compared to the classical design.

VIII. CONCLUSIONS

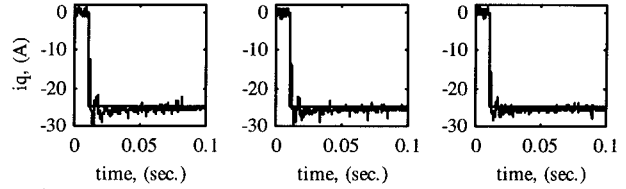
This paper has attempted to make two contributions: 1) development of complex vector methods for current regulator design and 2) development of a new alternative design which has been named the "complex vector, synchronous frame PI".

It was shown that the use complex vector root locus and frequency response functions greatly enables controls design.

From the paper's theoretical and experimental results several important conclusions can be reached:



$K_i/K_p = 0.5 \cdot \hat{r}_s' / \hat{L}_{\sigma s}$ $K_i/K_p = \hat{r}_s' / \hat{L}_{\sigma s}$ $K_i/K_p = 2 \cdot \hat{r}_s' / \hat{L}_{\sigma s}$
a) Classical Synchronous Frame PI (160 Hz)



$K_i/K_p = 0.5 \cdot \hat{r}_s' / \hat{L}_{\sigma s}$ $K_i/K_p = \hat{r}_s' / \hat{L}_{\sigma s}$ $K_i/K_p = 2 \cdot \hat{r}_s' / \hat{L}_{\sigma s}$
b) Complex Vector Synchronous Frame PI (160 Hz)

Figure 21: Experimental q Axis Current Step Response for an Induction Motor with Classical and Complex Vector Synchronous Frame PI Current Regulator (without Decoupling), 200 Hz bandwidth, for $f_e = 160$ Hz, Shown in the Synchronous Reference Frame

- The performance of the classical synchronous frame PI current regulator degrades as the synchronous frequency approaches the current regulator bandwidth
- The performance degradation occurs even if the controller zero is exactly equal to the physical (electrical) time constant. Synchronous frame PI tuning methods other than pole/zero cancellation will exhibit similar degradation as the synchronous frequency increases since the asymmetric complex vector root locus and FRF are intrinsic to its structure.
- A complex vector synchronous frame PI design was developed which implements a more desirable pole/zero cancellation. Even with incorrect parameter estimates it exhibits little dependency on the synchronous frequency.
- The additional computational requirements of the improved synchronous reference frame current regulators are minimal.

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